Ex/EE/PC/B/T/223/2025

B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER EXAM 2025

POWER SUPPLY SYSTEMS

Time: Three hours (50 marks for each part) Full Marks: 100

Use separate answer script for each part.

PART I

Answer **any five** questions.

Figures in the margin indicate full marks

1. a) What do you understand by pulverized fuel firing? Discuss about its advantages and disadvantages. 6+4=10
   b) Name one accessory in steam power plant that can reduce the problem of 'fly ash'. Discuss working principle of the equipment.

2. What are the desirable features of a peak load plant? 4+6
   Discuss why steam power plant is preferred to operated as base load plant but gas turbine plant is preferred to operate as peak load plant.

3. With a suitable schematic diagram, discuss working principle of a nuclear power plant having pressurized water reactor. 10

4.a) a) Explain the significance of the load curve. 2

b) The variation of the load over the 24 hours of a day for a certain area is as given below. Draw the load curve and load-duration curve. Calculate average load, load factor and demand factor. 8

Time	3AM-6AM	6AM-9AM	9AM – 12 noon	12 noon-3PM	3PM-6PM	6PM-9PM	9PM-12AM	12AM-3AM
Load (MW)	20	30	50	70	70	120	80	40

5. With the help of a schematic diagram discuss the working principle of a closed cycle gas turbine plant. 10

6. a) Write a comparative study among Pelton wheel, Francis turbine and Kaplan turbine. 6+4
   b) What are the factors to be considered in site selection of hydroelectric power plant?

7. Discuss the functions of (i) moderator in nuclear power plant (ii) compressor in gas turbine power plant (iii) surge tank in hydroelectric power plant (iv) superheater of steam power station (v) starting motor in gas turbine plant. 10

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PART – II

Group A

Answer *any three* of the following questions.

1. The DC distributor shown in Fig.Q.1 has a uniform load of 1A/m in addition to the concentrated load. The total resistance of the distributor is 0.075Ω, and it is fed at 230V at A. Determine the voltages at each load point. Derive the necessary formula. (6+4)

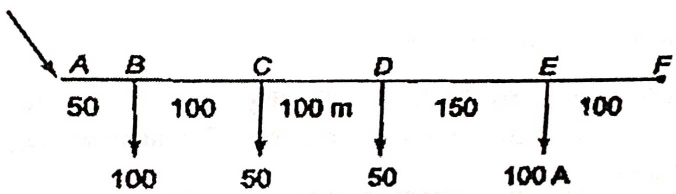

Fig.Q.1.

2. (i) Explain the basis for the design of distributors and feeders in the power supply system. (4)
(ii) Differentiate between radial system, ring main system and ring main with interconnector for a distribution system. (6)

3. (i) What are the limitations of Kelvin's law? (3)
(ii) Determine the most economical current density in a three phase 11kV cable which supplies 1500kW at 0.8 pf lagging for 300 days in a year at an average rate of 8 hours per day. The resistance per km length of the cable of 1sq. cm cross sectional area is 0.173ohm and the rate of interest and depreciation is 12%. Assume cost of energy loss as 2 paisa per unit and capital cost per km of the cable as Rs. (8000+20,000a) where a is the cross sectional area in sq. cm. (7)

4. (i) Explain why the maximum voltage between each conductor and earth forms the basis for comparison of volume of conductor for overhead transmission line and for underground cables maximum voltage between two conductors is considered. (3)

(ii) Assuming transmitted power, maximum voltage between conductors, length of transmission line, power factor and transmission efficiency remaining same for all cases, compare the volume of copper required in an underground scheme for (i) AC single phase 3 wire system and (ii) AC three phase three wire system. (7)

5. (i) A three-wire d.c distributor 600 m long is supplied at 500/250 V and is loaded as under:

Load between the positive terminal and the neutral wire: 60 A, 200 m from the positive terminal; 40 A, 360 m from the positive terminal.

Load between the neutral wire and the negative terminal: 20 A, 100 m from the negative terminal; 60 A, 260 m from the negative terminal and 15 A, 600 m from the negative terminal.

The resistance of each outer is 0.02 Ω per 100 metres, and the cross-section of the neutral wire is the same as the outer. Find the voltage across each load point. (7)

(ii) For the transmission of electrical power through a transmission line, show that with the increase in transmission line voltage, the volume of conductor material required decreases. (3)

Group B

Answer *any two* of the following questions

6. (i) Discuss how the substation can be classified. (6)

(ii) Mention the functions of the isolator and circuit breaker in a substation. (4)

7. (i) Discuss the principle of operation of the HRC fuse with a suitable diagram. (6)

(ii) Mention the advantages of neutral grounding. (4)

8. (i) What do you understand by pipe and plate earthing? (6)

(ii) What are the advantages of high tension service main connections? (4)